

Student creative thinking in East Asia: The role of extracurricular activities and supportive environments

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ABSTRACT

Creative thinking is increasingly recognized as a critical competency for academic and lifelong success. Drawing on data from 16,893 students in the Four Asian Tigers—Singapore, Hong Kong, China, South Korea, and Chinese Taipei—this study investigates the direct and indirect effects of extracurricular activities and supportive environments on East Asian students' creative thinking. Results from structural equation modeling reveals that in-school activities, particularly those emphasizing expression and STEM domains, positively predict creative thinking via enhanced intrinsic motivation, curiosity, and risk-taking propensity. In contrast, out-of-school activities in similar domains often are often negatively associated with creative thinking, attributable to commercialization, credentialism, and performance-driven design. School and instructional support demonstrates complex effects: while indirect benefits emerge via individual traits, direct associations are often negative, reflecting tensions between creativity-oriented pedagogy and test-centric, hierarchical schooling cultures, as well as the superficial implementation of creative instruction. Peer and family support consistently promote creative thinking both directly and through individual traits. Extending Amabile's intrinsic motivation hypothesis and Sternberg and Lubart's investment theory, this study highlights the importance of autonomy-supportive environments that prioritize open-ended, non-instrumental learning over assessment-driven constraints in fostering creative thinking.

1. Introduction

Creativity is widely recognized as a vital competency for success across academic and professional settings. Creative thinking, a core component of creativity, represents the ability to generate novel and useful ideas within a specific domain, and is fundamental to broader creative capacities (Amabile, 1983; Barbot et al., 2015; Sternberg & Lubart, 1999). Creative thinking serves as a cornerstone of innovation, problem-solving, and societal advancement. Primary and secondary school periods are considered essential for cultivating students' creative thinking skills rather than focusing solely on creative accomplishments or advanced competencies (Barbot et al., 2015). Existing evidence consistently supports the view that fostering creative thinking should be a central objective of creativity

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education during primary and secondary education (Runco, 2003), as it provides a foundation for future creative achievements (Torrance, 1972). As adolescents undergo significant cognitive, emotional, and social development, nurturing their creative thinking becomes paramount (van der Zanden et al., 2020). Therefore, it is important to identify effective strategies for fostering creative thinking during adolescence.

Exant empirical research predominantly focuses on the contributions of single factors to students' creative thinking, examining either individual traits—such as intrinsic motivation (Emami et al., 2023), curiosity (Evans & Jirout, 2023; Grossnickle, 2016; Gross et al., 2020), and risk-taking propensity (Amabile & Gryskiewicz, 1988; Lei et al., 2021)—or environmental factors, including school settings (Asigigan & Samur, 2021; Davies et al., 2013), family support (Blanco-Novoa et al., 2021; Jankowska & Karwowski, 2019; Zhang et al., 2024; Zhao & Yang, 2021), teacher and peer support (Hsia et al., 2021; Lin et al., 2024), as well as societal influences (Barai & Saha, 2022; Csikszentmihalyi, 2014). Building on the investment theory of creativity (Sternberg & Lubart, 1991, 1996), which emphasizes the importance of supportive and resource-rich environments for creative thinking development, international frameworks such as those advanced by the OECD (2024, 2025) have identified two core categories of environmental influences on adolescent creativity: extracurricular activities and supportive environments. Extracurricular activities refer to structured, voluntary, and adult-supervised engagements outside formal instructional time (Mahoney et al., 2005), often encompassing domains such as arts-related, expressive, and STEM-oriented activities (OECD, 2024, 2025). Supportive environments, in turn, are characterized by autonomy, collaboration, and open-ended learning, and are further subdivided into peer support, family support, and school and instructional support (Amabile, 1996; Hoferichter et al., 2021). This two-pronged framework has been widely integrated into educational policy across East Asia, including in Hong Kong, Singapore, and mainland China, where fostering creativity has become a key competency goal (Chen & Huang, 2017; Curriculum Development Council, 2017; Tan & Ng, 2021). Although adolescent creative thinking is increasingly understood as emerging from the dynamic interplay between individual traits and environmental contexts (Karwowski, 2014; Lubart, 2017), empirical investigations rarely examine these domains in an integrated manner. As a result, the relative contributions of individual traits, extracurricular activity involvement, and supportive environments remain insufficiently understood (Asigigan & Samur, 2021; Csikszentmihalyi, 2014). Moreover, while prior studies have explored how extracurricular activities or teacher support influence individual traits such as curiosity and intrinsic motivation (Jankowska & Karwowski, 2019; Zhao & Yang, 2021), limited research has explored whether these environmental factors indirectly shape students' creative thinking via their influence on personal traits.

East Asian societies, long noted for their strong academic performance on international assessments such as PISA and TIMSS, also achieved high rankings in creative thinking on the PISA 2022 assessment, challenging the assumption that rigorous academics inhibit creativity (OECD, 2022). This suggests that East Asian cultures may possess unique advantages or mechanisms for fostering creative thinking. However, most existing studies have focused on Western students (Kim, 2009), leaving the influence of these factors in East Asia underexplored. Mainland China, Singapore, Japan, Chinese Taipei, Hong Kong-China and Korea are widely regarded as prototypical East Asian societies, sharing Confucian values (Minkov, 2013). As mainland China and Japan did not participate in the PISA 2022 creative thinking assessment, this study focuses on the “Four Asian Tigers”—Chinese Taipei, Hong Kong, Singapore, and Korea—whose shared Confucian heritage and advanced economies provide a culturally representative and frequently studied context (Morris, 1996; Tu, 1996) for examining how environmental influences shape creative thinking.

While existing research has examined individual traits and environmental influences on adolescents' creative thinking, the interactive effects between these domains remain insufficiently explored—particularly in non-Western sociocultural contexts. Furthermore, although East Asian students consistently demonstrate high levels of academic achievement, the culturally situated mechanisms that foster creative thinking in these settings have received limited empirical attention. To address these gaps, the present study examines the “Four Asian Tigers” as representative East Asian societies. It investigates two interrelated research questions: (1) How do extracurricular activities and supportive environments influence students' creative thinking in East Asian contexts? (2) To what extent do these environmental factors contribute to the development of individual traits—such as curiosity, intrinsic motivation, and risk-taking—that are empirically linked to creative thinking? By integrating psychological and contextual perspectives, this study offers a culturally grounded understanding of how creative thinking develops and provides insights for designing educational systems that more effectively nurture creativity in East Asian contexts.

2. Literature review

2.1. From individual traits to environmental interactions: advancing theories of creative thinking in educational contexts

Research on creativity has historically been grounded in individual-centered models, such as Sternberg's (1988) triarchic theory and Amabile's (1988) componential theory, which emphasize cognitive ability, motivation, and domain-relevant skills. Empirical evidence has consistently shown that traits like intrinsic motivation, curiosity, and openness significantly predict creative thinking (Emami et al., 2023; Torrance, 2003). Over time, theoretical developments have moved beyond the traits to highlight individual-environment interactions. Sternberg and Lubart's (1991) investment theory underscores the alignment between individual potential and environmental support (Zhang & Sternberg, 2011). Beghetto and Kaufman's (2007) Mini-C theory focuses on early developmental expressions of creativity, while Glăveanu's (2013) 5A model conceptualizes creativity as a dynamic process shaped by sociocultural context (Zhang et al., 2023). This theoretical shift reflects a move from individual-centered views to interactionist perspectives, emphasizing creativity as a product of the dynamic interplay between personal potential and environmental support.

Although most creativity theories originated in Western contexts, East Asian perspectives contribute distinct conceptualizations. Western frameworks often frame creativity as divergence from social norms and individual originality (Gardner, 1996), whereas East

Asian paradigms emphasize harmony through dialectical integration as a driver of creative expression (Morris & Leung, 2010). Creativity in East Asia is embedded in collective practices, normative expectations, and institutional structures that contrast with Western individualistic assumptions. Building on this cultural distinction (Csikszentmihalyi, 2014), the present study investigates how individual traits and environmental factors jointly influence students' creative thinking within East Asian educational contexts.

2.2. Dimensions of supportive environments and extracurricular activities in East Asian educational contexts

Recent research has increasingly underscored the importance of contextual and environmental factors—particularly supportive environments and participation in extracurricular activities—in fostering student creativity (Davies et al., 2013; Mahoney et al., 2005; OECD, 2024, 2025). Despite cultural variability, the structural components of supportive environments—namely peer, family, and instructional support—exhibit substantial cross-cultural consistency (Amabile, 1996; Hoferichter et al., 2021). This tripartite framework has been validated across diverse educational settings (Deng et al., 2016; Zhang et al., 2021).

The organization and implementation of extracurricular activities reveal structural variation across cultural contexts. In Western contexts, creativity-oriented activities typically fall into three categories: school-based or external arts programs, expressive practices, and STEM-related initiatives (Egana-delSol, 2023; Marozzo et al., 2024; Samaniego et al., 2024). However, East Asian systems—especially those of the Four Asian Tigers—adopt a more institutionalized and multidimensional approach to extracurricular programming, aligning such activities with national objectives for 21st-century skills and creative development (Cheng, 2017; OECD, 2024). Despite their structural breadth, these programs broadly converge into three similar functional domains commonly associated with creativity: arts-related, expression-related, and STEM-related activities. For example, Hong Kong's Other Learning Experiences (OLE) framework includes domains such as arts development and community service, aligning with both arts-related and expression-related categories (Curriculum Development Council, 2017). In Singapore, creative thinking—termed “innovative thinking”—is integrated across curricula and strengthened through STEAM-based activities that fuse science, technology, engineering, the arts, and mathematics (Harris & de Bruin, 2017; Ministry of Education, Singapore, 2024). South Korea incorporates extracurricular domains such as the arts, humanities, and scientific inquiry into its formal curriculum (Ministry of Education, Republic of Korea, 2021), while Taipei emphasizes civic participation, technological innovation, and student leadership as part of both co-curricular and extracurricular learning (Chen & Huang, 2017). These extracurricular activities are often embedded within inquiry-based, collaborative learning environments supported by teacher facilitation (Lin & Tsai, 2015).

2.3. The direct effects of supportive environments and extracurricular activities on creative thinking

Based on a synthesis of existing research, supportive environments for creativity can be categorized into three major domains: peer, family, and school and instructional contexts. Peer support fosters a collaborative atmosphere conducive to risk-taking and idea-sharing, which are essential for creative development (Barai & Saha, 2022). In East Asian contexts, strong peer relationships have been linked to enhanced creative thinking through interactive mechanisms such as peer assessment and mutual feedback (Liu et al., 2023, 2024). Empirical evidence from both Western and Eastern contexts highlights the positive influence of family support—particularly autonomy granting and encouragement of exploration—on students' creative potential (Deng et al., 2016; Jankowska & Karwowski, 2019; Zhang et al., 2021). Parental involvement enhances creative self-efficacy, thereby facilitating creative behavior (Tang et al., 2022; Zhao & Yang, 2021). Within school settings, instructional environments that emphasize autonomy, non-authoritarian classroom climate, open-ended tasks, divergent teaching strategies, and supportive teacher–student relationships contribute positively to creativity (Hsia et al., 2021; Kupers & Dijk, 2020; So & Hu, 2019; Tan, 1998). Nonetheless, several studies highlight structural and cultural constraints within East Asian educational systems (Feng & Xiao, 2024). In Taipei, rigid instructional practices and limited acceptance of student exploration may hinder creative expression (Ting, 2013). Confucian-influenced norms emphasizing conformity and error avoidance can suppress risk-taking or divergent thinking, particularly in Korea, where teacher authority and exam-driven shadow education reduce students' creative engagement (Han & Suh, 2023; Kim et al., 2011). Similarly, in Hong Kong, traditional pedagogical attitudes have been shown to limit the impact of creativity-oriented interventions (Cheng, 2011).

Beyond supportive environments, creative extracurricular activities are essential pathways for cultivating students' creative thinking, typically categorized into three domains—arts-related, expression-based, and STEM-focused activities—conducted both in and outside of school. Engagement in the arts—such as visual arts and music—has been consistently linked to enhanced creative and academic outcomes, fostering abstract reasoning and adaptive thinking (Du et al., 2020; Egana-delSol, 2023; Samaniego et al., 2024). Expression-based activities—including drama, creative writing, and debate—similarly cultivate divergent thinking and narrative fluency (Samaniego et al., 2024). In Hong Kong, structured drama education has been shown to enhance students' creativity-related competencies (Hui & Lau, 2006), a finding echoed in Western contexts where drama promotes collaboration and creative problem-solving (Garaigordobil, 2011). STEM-based extracurriculars also contribute to creative development, particularly through digital learning environments that nurture “6C” skills including creativity and critical thinking (Marozzo et al., 2024). However, evidence from East Asian contexts suggests that rigid, exam-oriented implementation may constrain creativity. Korean students, for instance, reported that domestic STEM activities often emphasized correctness over innovation (Lee, 2017), while similar patterns were observed in the Philippines and Vietnam, where high-stakes pressures and insufficient pedagogical support discouraged risk-taking (Chauhan et al., 2021).

2.4. The indirect influence of supportive environments and extracurricular activities on creative thinking: mediating roles of intrinsic motivation, curiosity, and risk-taking propensity

In addition to external environmental influences, a growing body of research underscores the importance of individual traits—particularly intrinsic motivation, curiosity, and risk-taking propensity—in fostering students' creative thinking (Emami et al., 2023; Evans & Jirout, 2023; Grossnickle, 2016; Gross et al., 2020; Lei et al., 2021; Torrance, 2003). These traits not only directly shape creative outcomes but also function as mediators linking environmental factors to creative outcomes.

Intrinsic motivation, defined as the internal drive to engage in an activity for its own sake, is a well-established driver of creativity (Amabile, 1983). It enhances persistence, engagement, and originality, and has been found to mediate the relationship between environmental support and creative outcomes across diverse contexts (Emami et al., 2023; Zhang et al., 2021). Notably, studies in East Asia—such as Zhang et al. (2021)'s cross-regional analysis—demonstrate that creative motivation plays a significant mediating role between peer and family support and students' creative thinking, particularly under academically competitive conditions. Curiosity, defined as the recognition of knowledge gaps and the drive to explore, plays a pivotal role in creative thinking by prompting information-seeking behaviors (Grossnickle, 2016). While some empirical studies support a positive association between curiosity and creative thinking (Gross et al., 2020; Zhang et al., 2024), others, including recent behavioral research (Evans & Jirout, 2023), report non-significant correlations. These mixed findings suggest that curiosity may function more as a moderator or amplifier within broader creative processes rather than exerting a uniform direct effect. Risk-taking propensity, defined as the willingness to engage with uncertainty and pursue novel approaches, is widely recognized as a key contributor to creative performance (Ward & Kennedy, 2017). Empirical studies, from Amabile and Gryskiewicz's (1988) work with R&D professionals to more recent educational research (Lei et al., 2021), consistently indicate that environments supporting balanced risk-taking—characterized by psychological safety and structured freedom—significantly enhance students' creative thinking.

Emerging evidence suggests that creative environments and activities not only influence creative thinking directly but also shape key individual traits—namely, intrinsic motivation, curiosity, and risk-taking propensity. Educational (Davies et al., 2013) and family contexts (Blanco-Novoa et al., 2021), as well as both curricular and extracurricular experiences (Asigigan & Samur, 2021), have been found to foster intrinsic motivation. Similarly, peer-oriented and collaborative activities, such as debate and the arts, are positively associated with heightened curiosity (Zhang et al., 2024). Risk-taking propensity, in turn, acts as a psychological mediator, through which participation in such activities enhances creative engagement by reducing fear of failure and increasing openness to novel experiences (Vermont Department of Health, 2015).

While substantial evidence has linked environmental factors to creative development, comparatively limited studies have empirically examined how these effects are mediated by individual traits. This study therefore investigates the direct and indirect effects of creative support environments and extracurricular activities on East Asian students' creative thinking, with particular attention to the mediating roles of intrinsic motivation, curiosity, and risk-taking propensity. By doing so, it aims to deepen understanding of how external and internal mechanisms interact to foster creative thinking among East Asian students. (see Fig. 1).

3. Methodology

3.1. Participants

The data for this study were drawn from the 2022 Programme for International Student Assessment (PISA), a large-scale international assessment conducted by the Organisation for Economic Co-operation and Development (OECD). PISA 2022 evaluated 15-year-old students' competencies in mathematics, reading, and science and, for the first time, assessed their creative thinking. To evaluate how well students are prepared for the workforce and their overall development, PISA developed a series of questionnaires and tests to measure students' creative thinking.

To examine how environmental factors shape students' creative thinking, this study utilized data from 16,893 15-year-old students

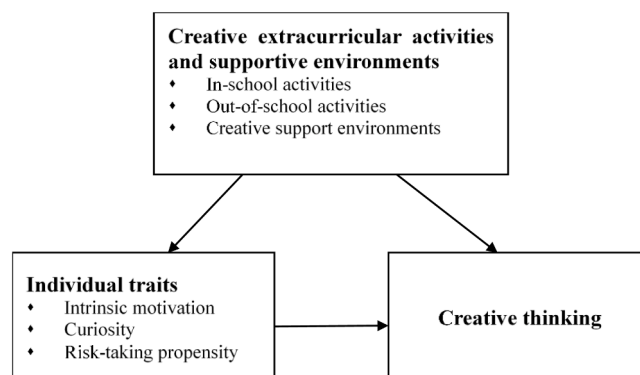


Fig. 1. The conceptual framework of the study.

in Chinese Taipei, Hong Kong-China, Singapore, and Korea. These economies—collectively known as the “Four Asian Tigers”—share a Confucian cultural tradition that emphasizes academic achievement, discipline, and deference to authority (Salili, 1996; Tu, 1996), alongside advanced levels of economic development and well-resourced, exam-driven education systems (Hanushek & Woessmann, 2023). This combination of cultural and institutional characteristics offers a theoretically rich and empirically relevant context for exploring creativity within high-performing, high-pressure educational environments (Kim, 2009; Morris, 1996).

3.2. Variables

3.2.1. Dependent variable

Creative thinking is defined as “the competence to engage productively in the generation, evaluation, and improvement of ideas that can result in original and effective solutions, advance knowledge, and impactful expressions of imagination” (OECD, 2022, p. 47). The PISA 2022 framework measures creative thinking through tests, including open-ended, computer-based, human-rated tasks. In these tasks, students demonstrate their creativity by responding to open-ended questions and practical scenarios. Students’ responses are assessed based on attributes such as originality, flexibility, fluency, and effectiveness, culminating in a standardized composite score ranging from 0 to 60.

3.2.2. Independent variables

Extracurricular activities and supportive environments were the independent variables in this study. In PISA 2022, eight extracurricular activities were identified as key predictors of students’ creative thinking (OECD, 2024). Drawing on prior research, these activities were categorized into three domains—arts-related, expression-related, and STEM-related—based on their participation modes and cognitive demands (Mahoney et al., 2003; Niu et al., 2022). **Arts-related activities** included engagement in “art classes / activities (e.g. painting, drawing),” “Music classes/activities (e.g. chorus, band),” and “Dramatics, theatre class/activities”) (in school: $\alpha = 0.750$; out of school: $\alpha = 0.848$). **Expression-related activities** were designed to foster creative expression and included options such as “Creative writing classes/activities,” “Debate [club]” and “Publications (e.g. newspaper, [yearbooks], literary magazine)” (in school: $\alpha = 0.750$; out of school: $\alpha = 0.848$). **STEM-related activities** involved science- and technology-based activities, including “Science [club]” and “Computer programming classes / activities” (in school: $\alpha = 0.740$; out of school: $\alpha = 0.786$). Students were asked to report the frequency of their participation in these activities, both in and outside of school. Responses were recorded on a five-point Likert scale, ranging from “1 = Never” to “5 = Every day or almost every day”.

Supportive environment captures the degree of creative thinking support provided by students’ peers, family, and teachers (Amabile, 1996; Hoferichter et al., 2021). **Peer support** was assessed through three items that measure openness and collaborative spirit among friends, including statements such as “My friends are open to new ideas,” “My friends and I give one another feedback about our ideas,” and “My friends and I encourage each other to come up with new ideas” ($\alpha=0.899$). **Family support** was evaluated through three items capturing the extent to which families foster curiosity and imagination, such as “My family encourages me to try new things,” “At home, I am encouraged to use my imagination,” and “Discussions I have at home help me come up with new ideas” ($\alpha=0.855$). **School and instructional support** was measured through six items reflecting the opportunities provided by teachers to express and value creativity. Statements include “My teachers give me enough time to come up with creative solutions on assignments,” “My teachers value students’ creativity,” “My mathematics assignments require me to come up with different solutions for a problem,” “My teachers encourage me to come up with original answers,” “The activities we do in my classes help me think about new ways to solve problems,” and “At school, I am given a chance to express my ideas,” ($\alpha=0.869$). Each item was rated on a four-point Likert scale ranging from “1 = Strongly disagree” to “4 = Strongly agree”.

3.2.3. Mediating variables

Intrinsic motivation, curiosity, and risk-taking propensity were potential mediators in this study. **Intrinsic motivation** refers to the inner drive to engage in tasks out of personal satisfaction and interest in creative problem-solving. It was measured using items in which students rated their agreement with statements such as “I like creating stories,” “I enjoy thinking about new ways to solve problems,” “I like games that challenge my creativity,” “I enjoy projects that require creative solutions,” and “I enjoy solving complex problems” ($\alpha = 0.868$). **Curiosity**, defined as the desire to explore new ideas and understand how things work, was assessed using items such as “I like to know how things work,” “I am more curious than most people I know,” and “I love learning new things in school” ($\alpha = 0.869$). **Risk-taking propensity**, referring to students’ tendencies to approach tasks either cautiously or impulsively, was measured by the difference between two sub-dimensions: caution and impulsiveness. The caution sub-dimension reflects a careful and meticulous approach to tasks, as captured by items like “I am careful with what I say to others,” “I like to make sure there are no mistakes,” “I carefully check homework before turning it in,” and “I think carefully before doing something” ($\alpha = 0.815$). Conversely, the impulsiveness sub-dimension indicates a tendency toward spontaneity and haste, as illustrated by items such as “I get easily distracted,” “I say the first thing that comes to my mind,” “I stop to think before acting,” and “I rush into activities without thinking” ($\alpha=0.755$). Each item was rated on a four-point Likert scale ranging from “1= Strongly disagree” to “4 = Strongly agree”.

3.2.4. Controlling variables

To account for potential confounders influencing creative thinking, the analysis included controls at the individual, family, and school levels. Individual-level controls consist of student gender (0 = female, 1 = male) and academic performance, calculated as the average score across mathematics, science, and reading assessments (Baer & Kaufman, 2008). Family-level controls incorporated the Economic, Social, and Cultural Status (ESCS) index, a composite measure based on three indicators: highest parental occupation status,

highest level of parental education in years, and a measure of household possessions (HOMEPOS) as a proxy for family resources (OECD, 2022). School-level covariates included the average ESCS of the school's student demographic composition (Hanushek & Woessmann, 2023).

3.3. Data analyses

Structural equation modeling (SEM) with Mplus 8 (Muthén & Muthén, 2012) was used to examine the relationships between supportive environment, extracurricular activities, and creative thinking, with three individual trait variables as potential mediators. To address the research questions, baseline models were first fitted to assess the direct effects of supportive environments and extracurricular activities (both in and out of school) on students' creative thinking. After establishing these direct relationships, the three individual traits—*intrinsic motivation*, *curiosity*, and *risk-taking propensity*—were introduced into the path model to test their mediating effects. To assess mediation, indirect effects were estimated using the delta method standard errors, as recommended for confirming mediation effects (MacKinnon et al., 2002). Maximum likelihood estimation (MLR) was employed to account for non-normality in the data. Covariates, including students' academic performance, gender, ESCS, and average ESCS of their schools, were controlled for in all regression models of the SEM. Model fit was evaluated using several indices, including comparative fit index (CFI), Tucker-Lewis index (TLI), root mean square error of approximation (RMSEA), and standardized root mean square residual (SRMR). For continuous data, model fit was considered acceptable when CFI and TLI values exceeded 0.90, and RMSEA and SRMR values were below 0.06 and 0.08, respectively (Schreiber et al., 2006).

4. Results

4.1. Descriptive statistics

The descriptive statistics presented in Table 1 indicate comparatively low levels of participation in creative extracurricular activities outside of school among 15-year-old students in the “Four Asian Tigers” regions. The mean participation for out-of-school art-related activities is 1.65 (SD = 0.94), with lower means for expression-related activities at 1.40 (SD = 0.81) and STEM-related activities at 1.41 (SD = 0.86). In contrast, school-based participation is higher: art-related activities have an average score of 2.17 (SD = 1.03), followed by expression-related at 1.81 (SD = 0.91) and STEM activities at 1.82 (SD = 1.05). Students reported relatively high levels of support from peers and family, with mean scores of 2.96 (SD = 0.55) and 2.94 (SD = 0.59), respectively. Support from school and instructional sources was slightly lower, with a mean score of 2.81 (SD = 0.55).

As shown in Table 1, intrinsic motivation has a slightly negative mean of -0.04 (SD = 1.00), while curiosity exhibits a marginally positive mean of 0.04 (SD = 1.00). Students display a relatively cautious disposition, as reflected in a higher average score for risk-taking propensity at 0.67 (SD = 1.31). Notably, students demonstrate elevated levels of creative thinking, with a mean score of 36.98 (SD = 10.75), which exceeds the OECD average of 33. This highlights the considerable creative potential among students in these regions.

4.2. Direct effects of extracurricular activities and supportive environments

Fig. 2 depicts the significant standardized paths from different types of extracurricular activities and supportive environments to students' creative thinking and individual traits, including *intrinsic motivation*, *curiosity*, and *risk-taking propensity*. In terms of

Table 1
Descriptive statistics of the full sample.

Variable	N	Mean	Std. Dev.	Min	Max
Out-of-school extracurricular activities					
Art-related	16,893	1.65	0.94	1	5
Expression-related	16,893	1.40	0.81	1	5
STEM-related	16,893	1.41	0.86	1	5
In-school extracurricular activities					
Art-related	16,893	2.17	1.03	1	5
Expression-related	16,893	1.81	0.91	1	5
STEM-related	16,893	1.82	1.05	1	5
Supportive environment					
School and instructional support	16,893	2.81	0.55	1	4
Peer support	16,893	2.96	0.55	1	4
Family support	16,893	2.94	0.59	1	4
Individual traits					
Intrinsic motivation	16,893	-0.04	1.00	-2.94	2.93
Curiosity	16,893	0.04	1.00	-4.97	4.31
Risk-taking propensity	16,534	0.67	1.31	-4	4
Creative thinking	16,893	36.98	10.75	1	58.10

Note. Data were unweighted.

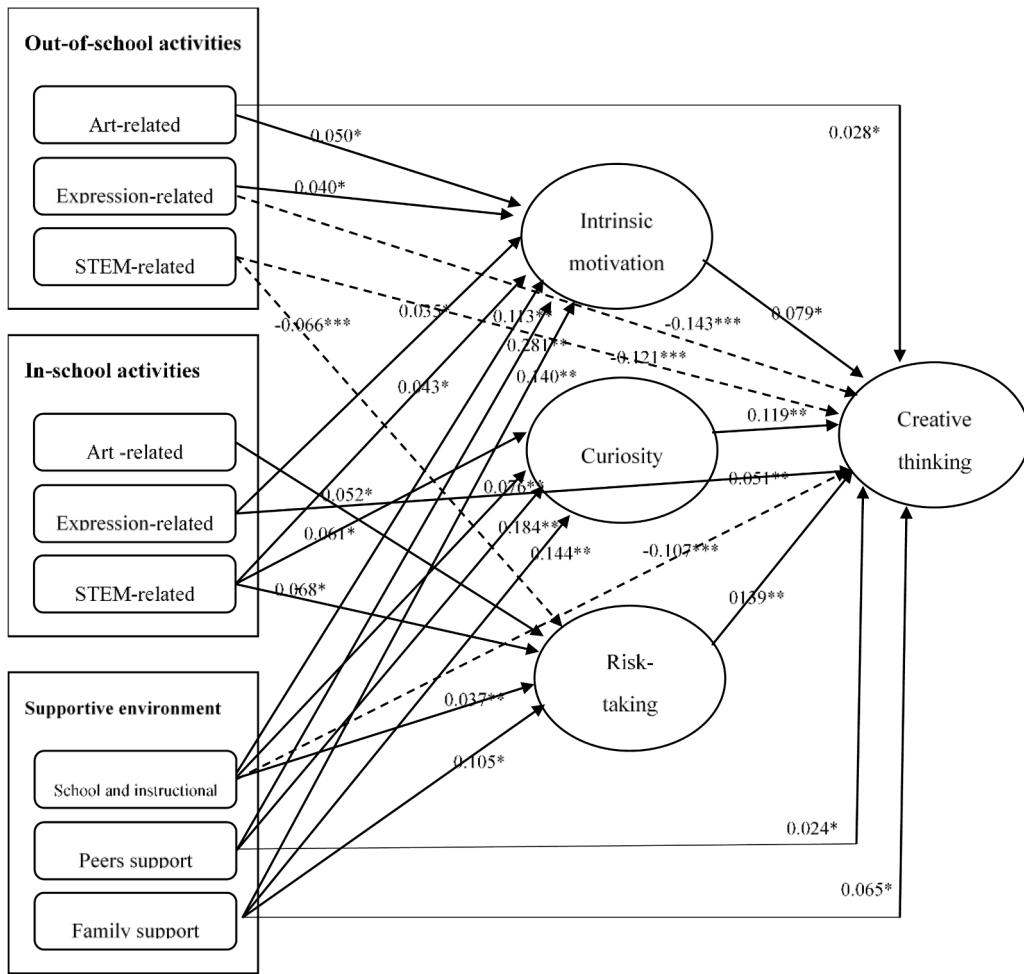


Fig. 2. Potential paths on students' creative thinking

Note. Standardized path coefficients are displayed. Solid lines represent positive relationships, while dashed lines indicate negative relationships between variables. For simplicity, results for covariates and correlations are omitted. Model fit indices are as follows: CFI = 0.954, TLI = 0.929, RMSEA = 0.046, SRMR = 0.016. * $p < .05$, ** $p < .01$, *** $p < .001$.

extracurricular activities, notable variations are observed in their influence on students' creative thinking. Specifically, in-school expression-related activities exhibit a significant positive direct effect on creative thinking ($\beta = 0.08$, $p < .001$), whereas in-school art-related activities ($\beta = -0.001$, $p > .05$) and STEM-related activities ($\beta = 0.012$, $p > .05$) show no direct effects. In contrast, out-of-school activities demonstrate divergent effects. Out-of-school expression-related ($\beta = -0.145$, $p < .001$) and STEM-related activities ($\beta = -0.121$, $p < .001$) have significant negative impacts on creative thinking, while out-of-school art-related activities positively influence creative thinking ($\beta = 0.028$, $p < .05$).

Regarding supportive environments, differential effects are also evident. Peer support ($\beta = 0.024$, $p < .05$) and family support ($\beta = 0.065$, $p < .001$) positively contribute to students' creative thinking. However, school and instructional support has a significant negative direct effect on creative thinking ($\beta = -0.107$, $p < .001$).

4.3. Indirect effects of extracurricular activities and supportive environments

Extracurricular activities and supportive environments exert complex influences on creative thinking, mediated by intrinsic motivation, curiosity, and risk-taking propensity (Table 2). For out-of-school activities, art-related ($\beta = 0.004$, $p < .001$) and expression-related activities ($\beta = 0.003$, $p < .01$) exhibit significant positive indirect effects on creative thinking via intrinsic motivation. However, no mediation effects are identified through curiosity or risk-taking propensity, resulting in no significant total indirect effect on creative thinking (art: $\beta = 0.003$, $p > .05$; expression: $\beta = 0.003$, $p > .05$). In contrast, STEM-related activities show a negative indirect effect through risk-taking propensity ($\beta = -0.009$, $p < .001$), while failing to demonstrate mediation through intrinsic motivation or curiosity. This pattern produces a negative total indirect effect on creative thinking ($\beta = -0.007$, $p < .05$).

In-school activities, in comparison, yield more consistently positive outcomes. art activities enhance creative thinking indirectly

Table 2

The direct and indirect effects of extracurricular activities and support environments on students' creative thinking.

		Direct effects	Access to creative thinking (indirect)			
			Intrinsic motivation	Curiosity	Risk-taking propensity	Total indirect
Out-of-school extracurricular activities	Art-related	0.028* (0.012)	0.004*** (0.001)	0.001 (0.001)	−0.002 (0.002)	0.003 (0.003)
	Expression-related	−0.145*** (0.001)	0.003** (0.001)	−0.002 (0.002)	0.002 (0.002)	0.003 (0.003)
	STEM-related	−0.121*** (0.001)	0.001 (0.001)	0.001 (0.002)	−0.009*** (0.002)	−0.007* (0.003)
In-school extracurricular activities	Art-related	−0.001 (0.001)	−0.001 (0.001)	−0.002 (0.001)	0.007*** (0.002)	0.004 (0.002)
	Expression-related	0.051*** (0.007)	0.003** (0.001)	0.002 (0.001)	0.003 (0.002)	0.008** (0.003)
	STEM-related	0.012 (0.007)	0.003*** (0.001)	0.007*** (0.001)	0.009*** (0.002)	0.020*** (0.003)
Creative supportive environment	School and instructional support	−0.107*** (0.010)	0.009*** (0.001)	0.009*** (0.002)	0.005** (0.002)	0.023*** (0.003)
	Peer support	0.024* (0.011)	0.022*** (0.003)	0.022*** (0.002)	0.001 (0.001)	0.045*** (0.003)
	Family support	0.065*** (0.020)	0.011*** (0.002)	0.017*** (0.002)	0.015*** (0.002)	0.043*** (0.003)

Note. Standard errors in parentheses. *** $p < .001$, ** $p < .01$, * $p < .05$.

through risk-taking propensity ($\beta = 0.007$, $p < .001$), but they fail to achieve a significant total indirect effect ($\beta = 0.004$, $p > .05$) due to the absence of effects via intrinsic motivation or curiosity. Conversely, expression activities exhibit a significant total indirect effect on creative thinking ($\beta = 0.008$, $p < .001$), primarily mediated by intrinsic motivation ($\beta = 0.003$, $p < .01$). Similarly, STEM activities indirectly contribute to creative thinking through intrinsic motivation ($\beta = 0.003$, $p < .001$), curiosity ($\beta = 0.007$, $p < .001$), and risk-taking propensity ($\beta = 0.009$, $p < .001$), culminating in a robust total indirect effect ($\beta = 0.02$, $p < .001$).

Supportive environments also demonstrate significant mediation effects on creative thinking through individual traits. Support provided by school teachers ($\beta = 0.023$, $p < .001$), peers ($\beta = 0.045$, $p < .001$), and family ($\beta = 0.045$, $p < .001$) collectively enhances creative thinking indirectly. Among individual traits, intrinsic motivation emerges as a critical mediator, linking school and instructional support ($\beta = 0.009$, $p < .001$), peer support ($\beta = 0.022$, $p < .001$), and family support ($\beta = 0.011$, $p < .001$) to creative thinking. Similarly, curiosity mediates the relationship between creative thinking and support from schools ($\beta = 0.009$, $p < .001$), peers ($\beta = 0.022$, $p < .001$), and family ($\beta = 0.011$, $p < .001$). Risk-taking propensity further mediates the influence of teachers ($\beta = 0.005$, $p < .001$) and family ($\beta = 0.015$, $p < .001$) on creative thinking.

5. Discussion

This study examined the direct effects of creative extracurricular activities and supportive environments on East Asian students' creative thinking, as well as the indirect effects mediated by intrinsic motivation, curiosity, and risk-taking propensity. Our findings contribute to the creativity literature in three key respects. **First**, it advances theoretical models such as Amabile's (1983) intrinsic motivation hypothesis and Sternberg and Lubart's (1991) investment theory by emphasizing the importance of context–trait interactions in the development of creative thinking. Specifically, environments that support autonomy, curiosity, and risk-taking—whether in school, family, or peer settings—function as “greenhouses” that foster creative potential. Our findings underscore the emergent and dynamic nature of creative thinking, shaped by both individual traits and environmental affordances. **Second**, this study provides a novel explanation for the relatively high creative thinking scores of students from the Four Asian Tigers (Singapore, Hong Kong-China, South Korea, and Chinese Taipei) compared to OECD averages. While prior assumptions have attributed these outcomes to cultural factors or rhetorical policy commitments, our results suggest that the implementation of creativity-supportive environments and structured activities plays a more substantial role. In-school expression-related and STEM activities, as well as both in-school and out-of-school arts activities, show positive effects on creativity through their alignment with motivational pathways. This suggests that when creativity is embedded within institutional design—rather than left to informal or extracurricular margins—it can flourish even within high-pressure academic contexts. **Third**, the study highlights the analytical value of distinguishing between in-school and out-of-school extracurricular activities. While prior research and large-scale assessments such as PISA have treated extracurricular engagement as a unified construct, our disaggregated approach reveals important contextual differences. Specifically, expression- and STEM-related activities conducted outside school tend to align more closely with academic credentialism and test preparation, thereby suppressing intrinsic motivation and creativity. In contrast, school-based programs are often intentionally designed to alleviate academic pressure and promote exploratory learning. This distinction provides a theoretical refinement to the study of creativity and calls for more nuanced measurement frameworks in international assessments. These contributions are further illuminated by examining how different types of extracurricular activities function across varied educational contexts.

The divergent effects of in-school and out-of-school extracurricular activities in the Four Asian Tigers—Singapore, Hong Kong-China, South Korea, and Chinese Taipei—can be understood through the lens of educational reforms and sociocultural dynamics.

Recent decades have seen these regions gradually shift from rigid, examination-driven systems toward more holistic educational paradigms that emphasize creativity, socio-emotional learning, and student agency (Cheng, 2017; OECD, 2024). Within this framework, in-school extracurricular activities—including Singapore's co-curricular programs, Korea's after-school initiatives, and structured school-based activities in Hong Kong and Taipei—are often explicitly designed to alleviate academic pressure and promote soft skills. These activities, particularly those centered on self-expression and interdisciplinary STEM, have demonstrated significant positive associations with creative thinking, often mediated by increased intrinsic motivation, curiosity, and risk-taking behavior (Harris & de Bruin, 2017; Lin & Tsai, 2015). The school context provides a relatively low-stakes, autonomy-supportive environment in line with Amabile's (1983) intrinsic motivation principle, which posits that creativity thrives when learners feel empowered to explore without fear of failure.

In contrast, the effects of out-of-school extracurricular activities—especially in expression-related and STEM domains—are markedly more complex and, in several cases, negatively associated with creative thinking. While some out-of-school art-based activities continue to show beneficial effects (Du et al., 2020; Egana-delSol, 2023), expression- and STEM-focused programs tied to academic subjects often undermine creativity (Chauhan et al., 2021; Lee, 2017). This trend can be attributed to three interrelated factors: commercialization, credentialism, and performance-oriented design. First, commercialization of the out-of-school education sector—fueled by intense academic competition and parental demand for measurable outcomes—has led to a proliferation of fee-based programs emphasizing standardized deliverables such as coding certifications, writing contests, and exam preparation (Lee et al., 2023; Tan, 2022). These programs frequently prioritize efficiency and correctness over open-ended inquiry, discouraging experimentation and risk-taking—key drivers of creative development (Sternberg & Lubart, 1996). Second, credentialism transforms expressive and creative practices into strategic tools for academic advancement. For example, out-of-school writing activities often focus on producing publishable outputs for inclusion in university application portfolios rather than nurturing original voice or divergent thinking (Hendriks-Kim, 2021). The result is a templated, formulaic approach to “creative” production that suppresses genuine exploration in favor of external validation. Third, the performance-oriented design of many out-of-school STEM activities—such as coding boot camps or math Olympiad prep—emphasizes correct solutions, time efficiency, and competition, leaving little room for the iterative, uncertain processes essential to creative problem-solving (Chauhan et al., 2021; Lee et al., 2023). Even in expression-based activities, the framing of creativity as a means to academic ends rather than as an end in itself constrains the breadth and authenticity of student engagement. As a result, many out-of-school activities prioritize academic credentialing and exam preparation over exploratory learning, thereby constraining opportunities for creative development. These findings highlight the need to align extracurricular programming with students' psychological and developmental needs. While in-school activities benefit from structured, autonomy-supportive environments conducive to intrinsic motivation and creative risk-taking, out-of-school programs—particularly those linked to high-stakes academic domains—require systemic reform. Reducing performance pressures and reorienting program design toward open-ended inquiry are essential steps toward fostering creative thinking. This alignment is particularly urgent in educational systems where success remains narrowly defined by standardized achievement metrics.

Contrary to prior research (Kupers & Dijk, 2020; So & Hu, 2019), our findings suggest that school and instructional support may have a directly negative impact on students' creative thinking in East Asian contexts, aligning with observations by Cheng (2011) and Han and Suh (2023). This paradox can be explained by three interrelated factors. First, the performance-oriented nature of schooling—characterized by standardized testing, rigid curricula, and high-stakes assessments—frames instructional support primarily as a means to improve test outcomes rather than to encourage intellectual exploration. Consequently, students may perceive creative pedagogies as conflicting with academic success, leading to reduced engagement and intrinsic motivation (Beghetto & Kaufman, 2007). Second, cultural norms rooted in Confucian traditions, which emphasize conformity, hierarchy, and error avoidance (Han & Suh, 2023; Kim et al., 2011; Ting, 2013), shape teacher-student interactions toward directive rather than facilitative approaches. Such an environment suppresses risk-taking and divergent thinking, causing students to internalize a compliance-oriented mindset that limits creative expression. Third, creativity-focused instruction is often implemented superficially due to insufficient teacher training and weak institutional incentives. Teachers tend to rely on isolated or token creative activities without embedding them in a comprehensive, autonomy-supportive pedagogical framework (Feng & Xiao, 2024; Tan, 1998), thereby failing to provide the psychological safety and cognitive stimulation necessary for sustained creative engagement. Despite these constraints on direct effects, school and instructional support may still foster creative thinking indirectly by enhancing intrinsic motivation, curiosity, and risk-taking. These findings underscore the urgent need for instructional reforms that reduce test-centric pressures and build teacher capacity through sustained professional development to effectively support creativity in education.

Peer and family support demonstrate consistently positive direct effects on creative thinking, along with significant indirect effects via intrinsic motivation, curiosity, and risk-taking propensity. These findings align with previous studies (Liu et al., 2023; Tang et al., 2022; Zhao & Yang, 2021). Peer support fosters collaborative, non-judgmental, and low-risk environment conducive to exploration and idea generation (Lin et al., 2024). Similarly, family support—particularly when emphasizing emotional encouragement and valuing exploration over academic performance—enhances intrinsic motivation, facilitating engagement with creative tasks (Emami et al., 2023; Zhang et al., 2024). These mediating traits help students tolerate ambiguity, persist through challenges, and pursue novel ideas (Emami et al., 2023; Lei et al., 2021). In East Asian settings, cautious risk-taking—often rooted in cultural norms—may further promote thoughtful, nuanced creativity (Kaufman & Sternberg, 2010). These findings underscore the importance of cultivating socially supportive environments aligned with students' psychological needs and cultural contexts to foster creativity.

Despite its contributions, this study has several limitations. **First**, its focus on the Four Asian Tigers—Chinese Taipei, Hong Kong, Singapore, and Korea—limits generalizability across East Asia. While these systems share Confucian cultural roots and strong academic performance, they do not capture the region's broader educational heterogeneity, particularly in the absence of data from mainland China and Japan. Furthermore, this study does not conduct within-region analyses to explore potential heterogeneity among

the four jurisdictions themselves. Future research should examine cross-region differences in greater depth, accounting for variations in educational policies, extracurricular structures, and creativity-related practices. **Second**, the study was unable to assess the quality of extracurricular activities, despite its likely influence on creative development. Critical factors such as instructional design, facilitator expertise, and institutional resources remain unmeasured due to limitations inherent in the PISA dataset. While relevant covariates were controlled for, unobserved variation in program implementation may bias variation in program. Future research should incorporate detailed, process-level data to better elucidate how program quality and contextual factors influence creative outcomes.

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Declaration of generative AI in scientific writing

During the preparation of this work the authors used Claude-3.5 in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRedit authorship contribution statement

Jiali Li: Writing – review & editing, Writing – original draft, Methodology, Formal analysis. **Qixing Fan:** Writing – review & editing, Data curation. **Kai Zhao:** Writing – review & editing, Supervision, Conceptualization. **Yanan Zhang:** Writing – review & editing. **Huiqing Liang:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that there exists no competing financial interest or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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